

Xiaoqi Wu

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Currently based in: London / Suzhou

Ecological & Environmental Data Analysis | Shanghai / Suzhou / Hangzhou

Education

Imperial College London MSc Computational Methods in Ecology and Evolution London 2025.09 - 2026.09

- **Curriculum:** Quantitative methods in ecology and evolutionary biology, including data science, **GIS**, genomics, and bioinformatics

Imperial College London BSc Biological Sciences London 2022.09 - 2025.07

- **Relevant Coursework:** Ecology and Evolution, Fundamentals of Ecology, Behavioural Ecology, Field Skills in Ecology, Global Change Biology, Biodiversity and Conservation Biology, Programming and Statistics

Internship & Engagement

Shanghai Sunherb Pharmaceutical Technology Co., Ltd. Analytical Engineer · Analytical Department Shanghai 2024.07 - 2024.09

- **Data Analysis:** Independently carried out sample pre-treatment, HPLC system operation, and data analysis under strict quality-control procedures, ensuring accuracy and reliability – building cross-domain quantitative data-analysis experience

Research & Project Experience

Evaluating Multimodal LLM Extraction for Vector Trait Curation: High Precision with Structure-Dependent Record Recovery London 2025.12 - 2026.08

Python, Gemma 4 31B IT (multimodal LLM), PyMuPDF, Docling, prompt engineering, JSON Schema validation, bipartite record-matching evaluation, pytest

- **Benchmark Construction:** Designed a weighted greedy diversity-selection algorithm to select 20 papers (spanning trait types, reporting formats, and data sources) from VecTraits' 40,000+-row, 334-paper raw export; built a 2,335-record manually curated development benchmark and a 431-record held-out test set; reduced the 91-column VecTraits schema to a 22-field extraction schema aligned with the MIREVTD reporting standard – a reusable quantitative QC framework applicable to ecological data curation
- **Multimodal Extraction Pipeline:** Built a multimodal evidence pipeline combining PyMuPDF full-text extraction, Docling table-structure parsing, and 150 DPI figure/table rendering, fed into Gemma 4 31B IT (a 256K-context multimodal LLM) via prompt engineering and JSON Schema-constrained output; iterated across 11 prompt versions, reaching a development-set conditional macro field F1 of 0.810
- **Rigorous Evaluation & Results:** Designed a bipartite record-matching evaluation framework (row-set, conditional-field, and end-to-end field metrics); evaluated the locked pipeline on the held-out test set, achieving row precision of 1.000 and conditional macro field F1 of 0.729 – confirming a "high-precision, structure-dependent recall" profile suited to human-in-the-loop curation

Modelling the Impact of Climate Change on *Aedes aegypti* Population Dynamics in Florida London 2025.03 - 2025.06

R, deSolve, ggplot2, ODEs, statistical validation (RMSE, Pearson, Spearman)

- **Model Adaptation & Validation:** Adapted a published tropical ODE compartment model to simulate *Aedes aegypti* population dynamics in Collier County, Florida under temperature/rainfall effects; validated against 53 weeks of surveillance data (RMSE=1.65, Pearson $r=0.29$, Spearman $\rho=0.30$)
- **Outcome:** Optimized time-lag ($\tau = -2$ weeks) and a rainfall scaling factor; proposed model improvements (anthropogenic water sources, spatial heterogeneity, uncertainty quantification)

Macroecology Analysis on Trochilidae: Testing Latitudinal Gradients and Ecological Rules London 2024.10 - 2024.12

R, ggplot2, caper, GLM, PGLS

- **Hypothesis Testing:** Tested the latitudinal diversity gradient (LDG), Bergmann's rule, and Rapoport's rule using AVONET data and a phylogenetic tree, applying GLM and PGLS to model species richness against body mass, range size, and latitude
- **Findings:** Results strongly supported the LDG; Bergmann's rule was significant under a naive GLM but lost significance once phylogenetic signal was controlled for (PGLS) – a result that demonstrates the methodological rigor of separating statistical correlation from phylogenetic confounding

Effect of Environmental Flower Colour Ratios on Insect Colour Preference Cape Town 2024.09 - 2024.10

Field experiment, R (t-test, Pearson correlation)

- **Experimental Design:** Designed a 5-colour pan-trap field experiment across 6 sites during a South Africa field course to study insect flower-visiting colour preferences relative to environmental flower colour ratios
- **Results:** Yellow, blue, and white traps all significantly attracted insects over baseline ($p < 0.001$); yellow capture rate was positively correlated with the proportion of yellow flowers in the environment ($r = 0.369$, $p = 0.027$)

Skills & Other

- **Data Analysis & Modelling:** R (ggplot2, dplyr, caper, ape, vegan, deSolve), Python, statistical inference (GLM/Poisson regression, PGLS, Wilcoxon signed-rank, t-tests, Pearson/Spearman correlation), ODE- and difference-equation-based population dynamics modelling
- **Ecology & Fieldwork:** field experimental design, biodiversity/environmental surveying, GIS-based spatial analysis, macroecology and phylogenetic comparative analysis (AVONET-scale trait data, PGLS)
- **Languages:** English (IELTS 7.5), Mandarin Chinese (native), French (A2) | **Interests:** piano, photography, skiing